

The jogging speeds of 100, 10, 1.0 and 0.1 inches per second can be adjusted by the Feed Rate Override keys. This gives an additional 10% to 200% control.

5.3.5 Calculator

The number in the calculator box can be transferred to the data entry line by pressing **[F3]** in **Edit** or **MDI** mode. This transfers the number from the calculator box to the **Edit** or **MDI** input buffer (enter a letter, X, Z, etc., for the command to use with the number from the calculator).

The highlighted **Triangle**, **Circular**, **Or Turning** and **Tapping** data can be transferred to load, add, subtract, multiply, or divide in the calculator by selecting the value and pressing **[F4]**.

Simple expressions can be entered into the calculator. For example $23*4-5.2+6/2$, is evaluated when **ENTER** is pressed and the result (89.8 in this case) is displayed in the calculator box.

5.4 Basic Programming

A typical CNC program has (3) parts:

1. **Preparation:** This portion of the program selects the work and tool offsets, spindle speed, selects the cutting tool, and turns on the coolant.
2. **Cutting:** This portion of the program defines the tool path, and feed rate for the cutting operation.
3. **Completion:** This portion of the program turns off the coolant, moves the tool to Z-Axis home, moves the tool to X-Axis home, turns off the spindle, and allows the part to be unloaded from the chuck and inspected.

This program makes a 0.100" (2.54 mm) deep face cut in a piece of material with Tool 1 along the X Axis from X = 2.1 to X = - 0.02 (negative 0.02 X-Axis over-travel makes sure the uncompensated tool cuts the whole face).



NOTE:

A program block can contain more than one G-code, as long as those G-codes are from different groups. You cannot place two G-codes from the same group in a program block. Also note that only one M-code per block is allowed.

```
%  
o40001 (BASIC PROGRAM) ;  
(G54 X0 is at the center of rotation) ;  
(Z0 is on face of the part) ;  
(T1 is an end face cutting tool) ;  
(BEGIN PREPARATION BLOCKS) ;
```

```

T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X2.1 Z0.1 (Rapid to 1st position) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G01 Z-0.1 F.01 (Linear feed) ;
X-0.02 (Linear feed) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z0.1 M09 (Rapid retract, coolant off) ;
G97 S500 (CSS off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

5.4.1 Preparation

These are the preparation code blocks in the sample program:

Preparation Code Block	Description
%	Denotes the beginning of a program written in a text editor.
O40001 (BASIC PROGRAM) ;	O40001 is the name of the program. Program naming convention follows the Onnnnn format: The letter “O”, or “o” followed by a 5-digit number.
(G54 X0 is at the center of rotation) ;	Comment
(Z0 is on face of the part) ;	Comment
(T1 is an end face cutting tool) ;	Comment
T101 (Select tool and offset 1) ;	T101 selects the tool, the offset 1, and commands the tool change to Tool 1.

Preparation Code Block	Description
G00 G18 G20 G40 G80 G99 (Safe startup) ;	This is referred to as a safe startup line. It is good machining practice to place this block of code after every tool change. G00 defines axis movement following it to be in Rapid Motion mode. G18 defines the cutting plane as the XZ plane. G20 defines the coordinate positioning to be in Inches. G40 cancels Cutter Compensation. G80 cancels any canned cycles. G99 puts the machine in Feed per Rev mode.
G50 S1000 (Limit spindle to 1000 RPM) ;	G50 limits the spindle to a max of 1000 RPM. S1000 is the spindle speed address. Using Snnnn address code, where nnnn is the desired spindle RPM value.
G97 S500 M03 (CSS off, Spindle on CW) ;	G97 cancels constant surface speed (CSS) making the S value a direct RPM of 500. S500 is the spindle speed address. Using Snnnn address code, where nnnn is the desired spindle RPM value. M03 turns on the spindle.  NOTE: <i>Lathes equipped with a gearbox, the control will not select high gear or low gear for you. You must use a M41 Low Gear or M42 High Gear on the line before the Snnnn code. Refer to M41 / M42 Low / High Gear Override for more information on these M-codes.</i>
G00 G54 X2.1 Z0.1 (Rapid to 1st position) ;	G00 defines axis movement following it to be in Rapid Motion mode. G54 defines the coordinate system to be centered on the Work Offset stored in G54 on the Offset display. X2.0 commands the X Axis to X = 2.0. Z0.1 commands the Z Axis to Z = 0.1.
M08 (Coolant on) ;	M08 turns on the coolant.
G96 S200 (CSS on) ;	G96 turns on CSS. S200 specifies a cutting speed of 200 ipm to be used along with the current diameter to calculate the correct RPM.

5.4.2 Cutting

These are the cutting code blocks in the sample program:

Cutting Code Block	Description
G01 Z-0.1 F.01 (Linear feed) ;	G01 defines axis movements after it to be in a straight line. Z-0.1 commands the Z Axis to Z = -0.1. G01 requires address code Fnnn.nnnn. F.01 specifies the feedrate for the motion is .0100" (.254 mm)/Rev.
X-0.02 (Linear feed) ;	X-0.02 commands the X Axis to X = -0.02.

5.4.3 Completion

These are the completion code blocks in the sample program:

Completion Code Block	Description
G00 Z0.1 M09 (Rapid retract, coolant off) ;	G00 commands the axis motion to be completed in Rapid Motion mode. Z0.1 commands the Z Axis to Z = 0.1. M09 turns off the coolant.
G97 S500 (CSS off) ;	G97 cancels constant surface speed (CSS) making the S value a direct RPM of 500. On machines with a gearbox, the control automatically selects high gear or low gear, based on the commanded spindle speed. S500 is the spindle speed address. Using Snnnn address code, where nnnn is the desired spindle RPM value.
G53 X0 (X home) ;	G53 defines axis movements after it to be with respect to the machine coordinate system. X0 commands the X Axis to move to X = 0.0 (X home).
G53 Z0 M05 (Z home, spindle off) ;	G53 defines axis movements after it to be with respect to the machine coordinate system. Z0 commands the Z Axis to move to Z = 0.0 (Z home). M05 turns off the spindle.
M30 (End program) ;	M30 ends the program and moves the cursor on the control to the top of the program.
%	Denotes the end of a program written in a text editor.